Labor Demand and Production

Labor II — Week 1

Teaching deck draft

Central question

- How do firms choose labor input in a static production setting?
- Which objects belong to substitution across inputs, and which belong to scale?
- Why do labor-demand elasticities matter for policy incidence and for the rest of Labor II?

Course reorientation

- Labor I centered workers, households, and worker-side policy.
- Labor II begins inside the firm: employment is now an input-demand object.
- Week 1 is deliberately static: no adjustment costs yet, no bargaining yet, no monopsony yet.

Labor I to Labor II bridge

- Labor I asked how workers respond to wages, taxes, and institutions.
- Week 1 asks where firms want to be after those same wedges change.
- Week 2 will ask why firms often do not get there immediately.

Production and the static firm problem

$$q = F(L, K; A) \qquad pF_L(L, K; A) = w$$

- Labor demand is derived from technology, product demand, and factor prices.
- The value of the marginal product benchmark is useful but not enough.
- We need cost minimization to separate substitution from scale.

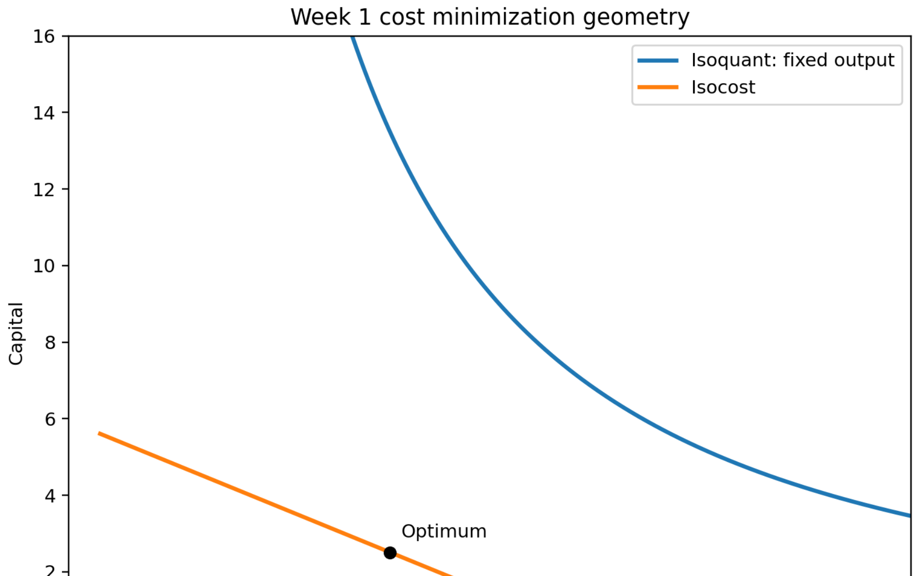
Cost minimization and conditional labor demand

$$\min_{L,K} \; wL + rK \quad \text{s.t.} \quad F(L,K;A) \geq q$$

$$\frac{F_L}{F_K} = \frac{w}{r} \qquad \eta^c_{LL} = -(1 - s_L)\sigma$$

- Conditional demand holds output fixed.
- It isolates substitution across inputs.
- It is not yet the full observed employment response to a labor-cost shock.

Isoquant and isocost geometry



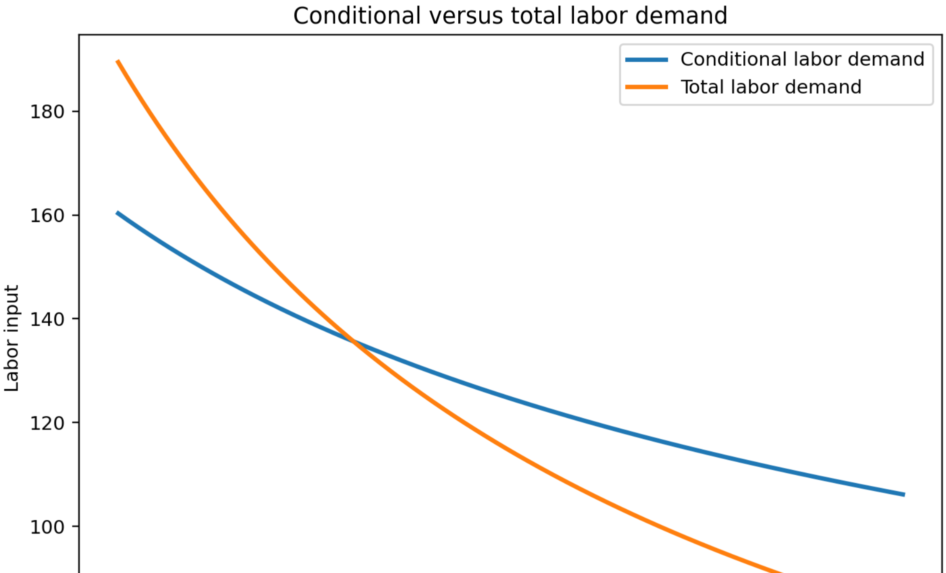
Unconditional labor demand and scale effects

$$\frac{d\ln L}{d\ln w} = \underbrace{\left.\frac{\partial \ln L^c}{\partial \ln w}\right|_q}_{\text{substitution}} + \underbrace{\frac{\partial \ln L}{\partial \ln q}\frac{d\ln q}{d\ln w}}_{\text{scale}}$$

$$\eta_{LL} = -(1 - s_L)\sigma - s_L\eta$$

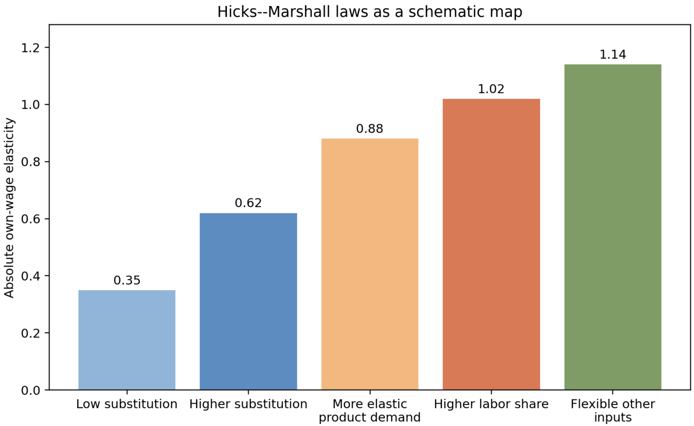
- Total labor demand combines input substitution with output contraction.
- Product-demand elasticity matters because cost shocks change desired scale.

Conditional versus total demand



Hicks–Marshall laws

- Labor demand is more elastic when substitution away from labor is easier.
- Labor demand is more elastic when product demand is more elastic.
- Labor demand is more elastic when labor is a larger share of cost.
- Labor demand is more elastic when other inputs are easier to expand.



Short run versus long run

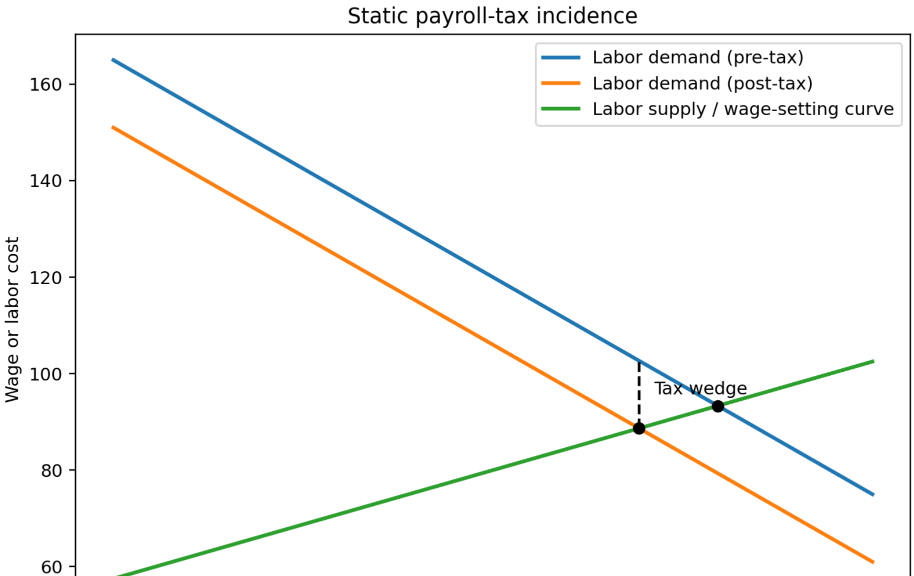
- Short run: capital, tasks, or organization are partly fixed.
- Long run: more substitution and reorganization margins are open.
- The theory is static, but the measured elasticity depends on the observed horizon.
- Week 2 will formalize the speed of adjustment.

Payroll taxes, cost shocks, and incidence

$$c_L = (1 + \tau)w$$

- Statutory incidence says who remits the tax.
- Economic incidence asks what shows up in wages, employment, prices, or profits.
- The benchmark employment response depends on total labor demand.

Payroll-tax wedge



Empirical design map

Design	Variation	Margin	Core object
Payroll-tax reform	Labor cost	Employment, wages	Total demand and incidence
City-industry shock	Local demand	Employment, wages	Demand with spillovers
Product-market cost shock	Marginal cost	Prices, employment	Product-demand link
Rent sharing	Firm rents	Wages, firm-worker outcomes	Boundary with wage-setting

Bridge to Week 2

- Week 1 tells us the firm's target after a shock.
- Week 2 asks why hiring and firing frictions slow movement toward that target.
- Dynamic adjustment costs turn a comparative-static destination into an employment path.

Takeaways

- Labor demand is derived from technology, costs, and product demand.
- Conditional demand is a substitution object; total demand adds scale.
- Hicks–Marshall logic disciplines policy incidence and empirical interpretation.
- Week 2 starts once we admit firms cannot jump to the new optimum costlessly.